

RAM CHARAN VANGA

RA2511027010164

Subject : DSA

Unit 2, S9, SLO1 - Circular Singly Linked List - Deletion

Algorithm to delete the first node

Step 1: IF START = NULL

Write UNDERFLOW

Go to Step 8

[END OF IF]

Step 2: SET PTR = _____

Step 3: Repeat Step 4 while PTR -> NEXT != _____

Step 4: SET PTR = PTR -> NEXT

[END OF LOOP]

Step 5: SET _____ = START -> NEXT

Step 6: FREE _____

Step 7: SET START = _____

Step 8: EXIT

Answer: Step 2: SET PTR = START

Step 3: while PTR -> NEXT != START

Step 4: SET PTR = PTR -> NEXT

Step 5: SET PTR -> NEXT = START -> NEXT

Step 6: FREE START

Step 7: SET START = PTR -> NEXT

Step 8: EXIT

Algorithm to delete the last node

Step 1: IF START = NULL

Write UNDERFLOW

Go to Step 8

[END OF IF]

Step 2: SET PTR = START

Step 3: Repeat Steps 4 and 5 while _____ != START

Step 4: SET _____ = PTR

Step 5: SET PTR = _____

[END OF LOOP]

Step 6: SET _____ = START

Step 7: FREE _____

Step 8: EXIT

Answer: Step 3: while PTR -> NEXT != START

Step 4: SET PREPTR = PTR

Step 5: SET PTR = PTR -> NEXT

Step 6: SET PREPTR -> NEXT = START

Step 7: FREE PTR

Step 8: EXIT